Project title: Saarthi: Optimizing Gig Worker Platform Recommendations using ML based on Environmental Dynamics and Worker Sentiment

Introduction:

- “Gig Economy” means a market in which they have fixed-term contracts or they pay per project arrangements where workers earn outside employer-employee relationship via digital SaaS platforms [1-2].
- Gig work provides new opportunities in developing countries with their huge populations. These countries face many challenges in providing jobs due to their huge population. Gig work platforms can help in overcoming unemployment problem by connecting abundant labor supply to market demand, while allowing organizations to significantly reduce permanent hiring costs [1].
- On the flip side, Gig workers lose various labour rights and legal protections, including wage guarantees, working hours restrictions, and working conditions [4-8]. The financial pressure of maintaining their own equipment (e.g., paying vehicle EMIs) often leads to poor mental health outcomes for workers [9].

Motivation:

- Decision Uncertainty: Gig workers make constant, time-sensitive decisions under uncertain conditions (changing demand, bad weather, traffic, fatigue).
- Gig workers depend on platforms for their earnings optimization and work allocation but platforms provide limited and isolated informations ,making it difficult to predict the platform that will give the best earnings/results for them .
- Core Problems Addressed
    - Unstable Earnings: Fluctuating income without predictive insights.
    - Health and Safety Risks: Avoidable burnout due to long hours and harsh environmental conditions.
    - High Mental Effort: Continuous stress from trial-and-error choices during daily operations.

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Existing work / Literature review:

- Growth and Precarity : Study by Roy and Shrivastava (2020) and Sharma and Sharma(2025) shows that the Gig work reduces unemployment in developing nations like India,Bangladesh through Software as a Service model ,it is done by removing employer-employee protections and wage guarantees .[1,2]
- Algorithmic Control and Labour Power : Research by Selcuk and Gokpinar(2026),Feng and Zhan (2019) show that flexible working conditions are governed by strict algorithmic control ,which forces workers to follow platform demand rather than their own schedules and reduces their overall bargaining power .[10,11]
- Mental Health and Burnout : Study by Wang(2022) and Gao et .at, (2025) shows that financial instability ,loneliness ,and constant supervision inherent in gig work lead to high levels of mental distress and burnout .[14]
- Financial apps like "KarmaLife" and "Entitled" provide tools to handle money for gig workers in India
- Insurance apps like “OnSurity”,” GIGI Benefits" provide/handles health coverage (insurance) for gig workers. Apps like “Vahan” and” Taskmo” help gig workers to find jobs in the market
- Platform applications like Zomato and Swiggy provide “Zomato Shelter”,” Swiggy Ambulance“ service, such as shelter (physical stops) and ambulances after an accident has happened for gig workers.
- **Gaps in the existing solutions/work**:
    - No single app does these functions together like what our platform does: Multi-platform recommendation, track worker health through sentiment analysis, burnout detection, and stress detection, and suggests apps based on weather context. and all done through voice, no forms needed to be filled in the process.

Work Done till date:

Details of proposed work/solutions:

- Voice Based Data Logging : Implement a voice module ,so that user interact using their voice and not fill the forms daily.
- Create data set : create a data set for our app to create ml models for predicting platform and burnout .
- Predict Platform : Develop an Ml algorithmto suggest the best possible platform for the worker for that day
- Predict Burnout : Develop an Ml algorithm that predicts workers Burnout.
- Build the full-stack system using Android ,backend , python backend
- Use external services which are required for our app.

Work Done till date:

Work done and results achieved:

- Implemented voice module using Google speech-to-text and Google text-to-speech Api that allows users to interact using voice .Daily earnings , Platform , Shift hours are extracted and collected through the transcription of that voice input .
- created a data set which is uniform , not basied using real weather and traffic data . and create synthetically workers work profile for the final data set .
- Developed an Ml algorithm(Xgboost-regression) using their earning data , weather,traffic data, sentiment data ,workers work data(experience,platform,age) to suggest the best possible platform for the worker for that day
- Developed an Ml algorithm that predicts workers Burnout using Xgboost multi class classification (high ,moderate,low).using the data set which was created.
- External services used : Google maps api,Google text to speech,Google speech to text,Open Weather api,Google gemini-2.5 for sentiment analysis .
- Achieved 90% R2 score for Platform recommendation Ml algo(Xgboost-regression) and Achived 92% accuracy for Burnout prediction Ml algo(Xgboost classification).

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